Code No: 118AA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
ADHOC AND SENSOR NETWORKS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July-Aug-2023

PART – A

(25 Marks)

1.a)	State any two challenges of Mobile Adhoc Networks (MANET).	[2]
b)	List the three packet forwarding schemes used in position-based routing.	[3]
c)	Mention the control packets used in ODMRP, a mesh-based protocol.	[2]
d)	What is meant by jitter in broadcast schemes for MANETS?	[3]
e)	State the relationship between communication and sensing ranges of a sensor node?	[2]
f)	Write the formula for next CH selection in LEACH protocol.	[3]
g)	State any two problems affecting secure ad hoc routing.	[2]
h)	What are the important modules in Intrusion Detection System (IDS) agent?	[3]
i)	What are main features of nesC language for building applications on TinyOS?	[2]
j)	Brief the mechanism of TinyGALS model.	[3]

PART – B

(50 Marks)

2.a)	Give the taxonomy of routing protocols for MANETS.	
b)	Distinguish the terms-Unicast, broadcast, multicast, geocast, and anycast.	[5+5]
	OR	
3.a)	Write about DREAM framework for location service.	
b)	Explain the greedy packet forwarding strategy.	[5+5]
4.a)	Illustrate the broadcast storm problem in MANETS.	
b)	Explain MGR, a geocast protocol, with a neat diagram.	[5+5]
	OR	
5.a)	Write the algorithmic steps for MPR, a Manet broadcast protocol.	
b)	Explain how MAODV works for multicast in MANETS.	[5+5]
6.a)	Giving an architectural diagram, write about Great Duck Island (GDI) application of WSNs.	
b)	What is meant by Directed Diffusion? How it works?	[5+5]
	OR	
7.a)	Explain how the TEEN protocol is suitable for time-critical sensing applications?	
b)	Illustrate the three stages of communication in SPIN protocol for adaptive routing.	[5+5]

8.a) Explain the important security risks of ad hoc networks.
b) Write about any one key agreement protocol explaining its notation. [5+5]

OR

9.a) What is meant by node cooperation in MANETS? Write one solution.
b) Give the architectural diagram of an Intrusion Detection System (IDS) for ad hoc networks. [5+5]

10.a) Explain the key features of NS2 simulator used for studying ad hoc networks.
b) What are the distinctions of TinyOS over conventional OS when studying concepts of ad hoc networks? [5+5]

OR

11.a) What is meant by a discrete-event simulator? Explain a tool based on it.
b) Explain the steps followed in developing an application using nesC language. [5+5]

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